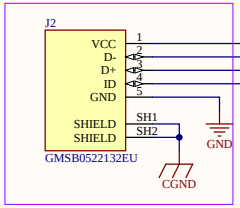
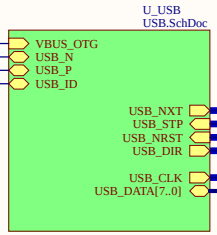
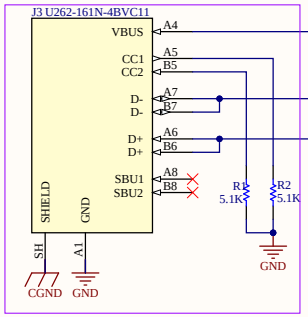
Zynq System Overview

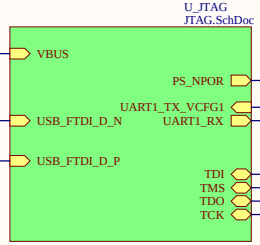
OTG Interface



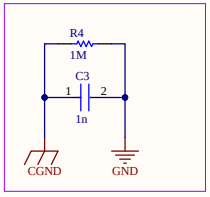


FTDI JTAG Interface

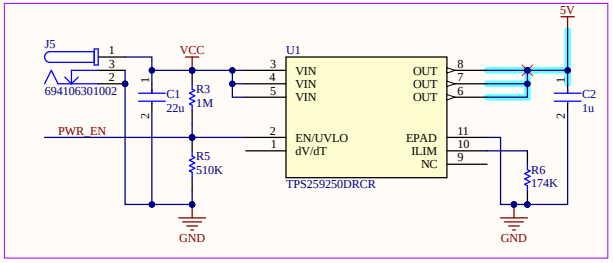


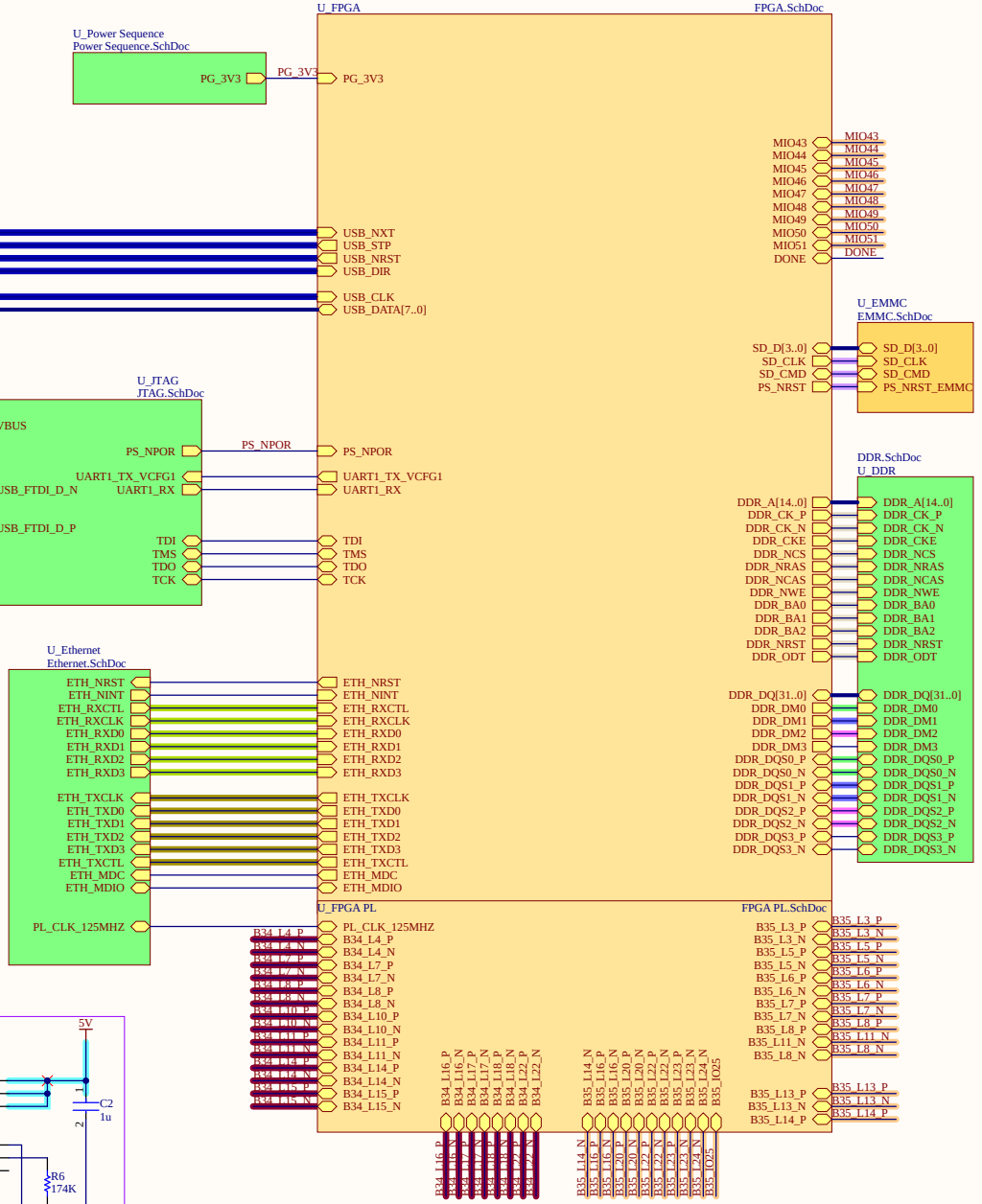


Chassis GND Connection

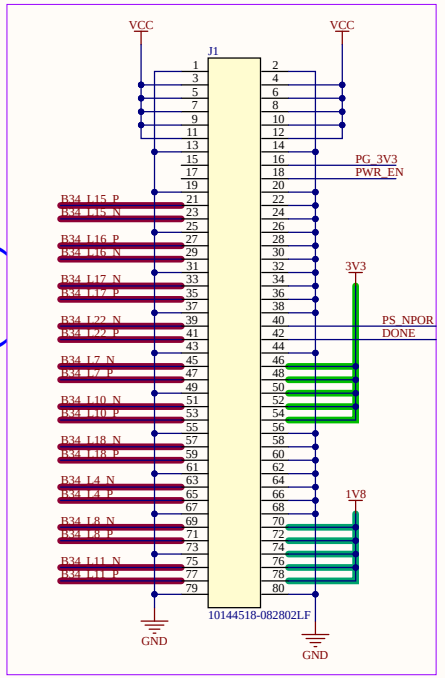


Input Power & eFuse

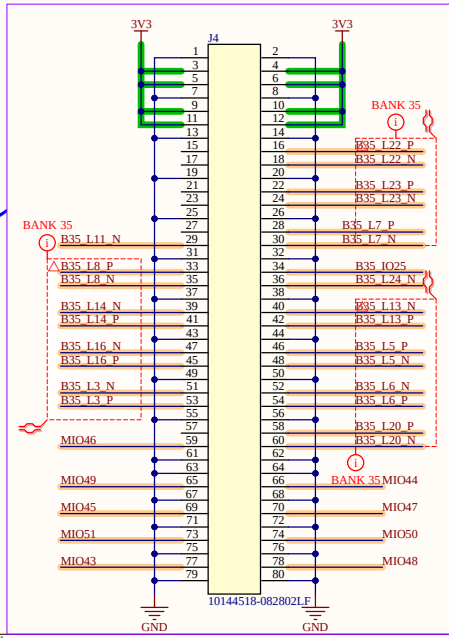




PL Bank 34 (DIFF)



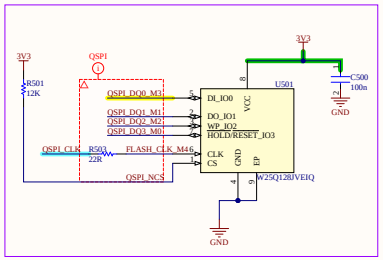
PL Bank 35 (SE



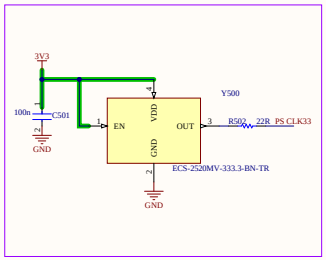
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Date: 10/27/2026		Sheet of
File: C:\Users\..\System.SchDoc		Drawn By:

Zynq Processing System and DDR Interface

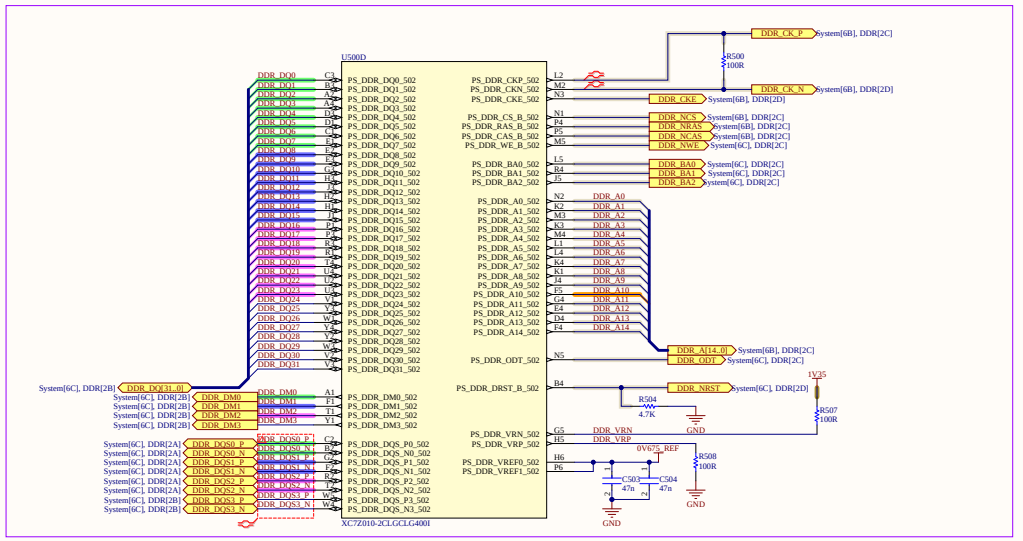
QSPIFlashMemory(128MBit)



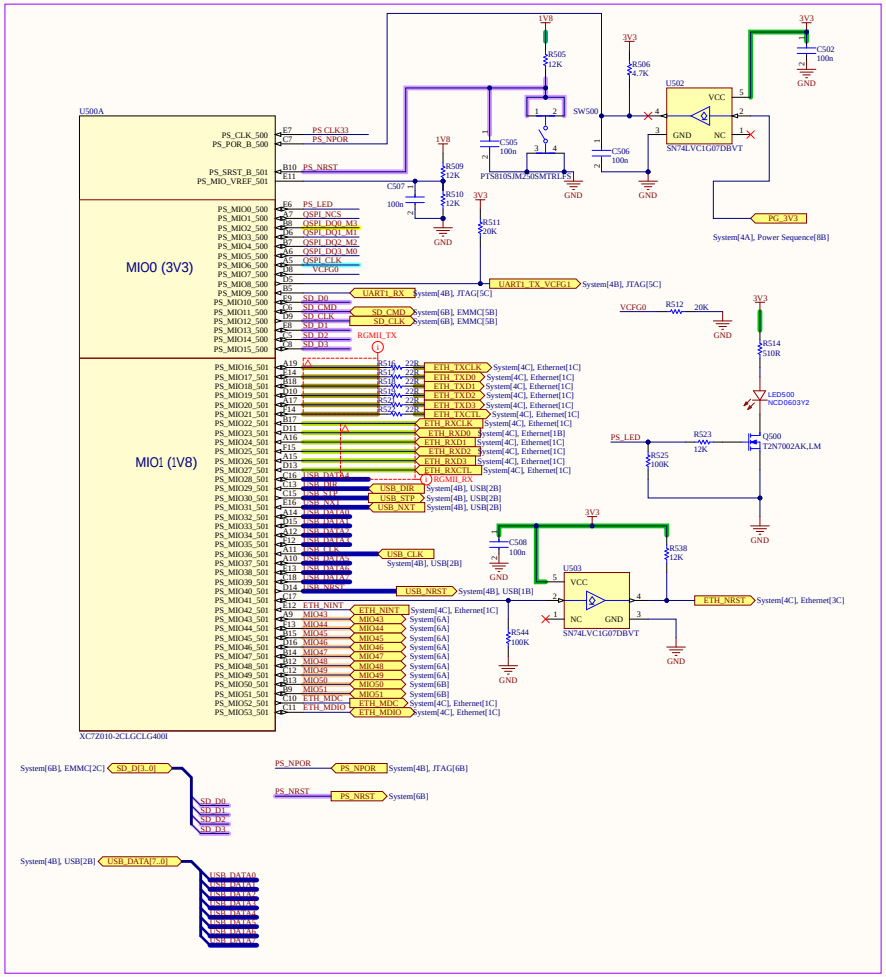
PSClock(33.33MHz)



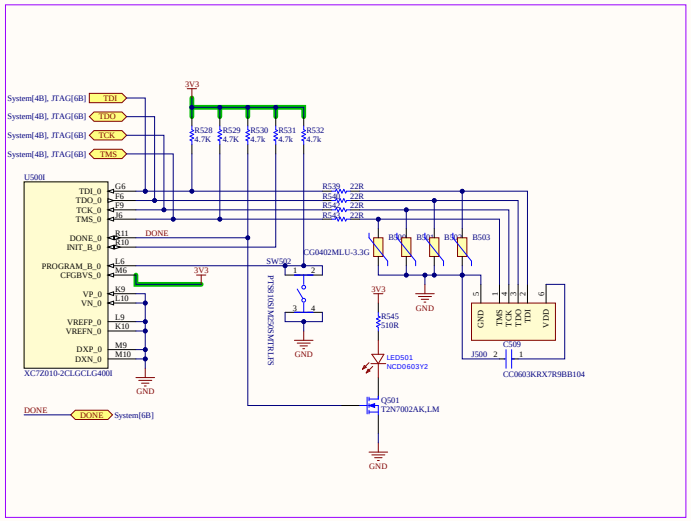
Zynq DDR Interface



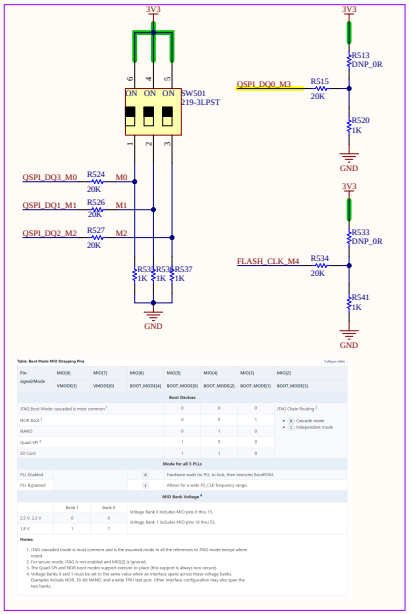
Zynq PS (Bank 500 & 501)



Zynq Config, JTAG



Boot Mode MIO Strapping Pins



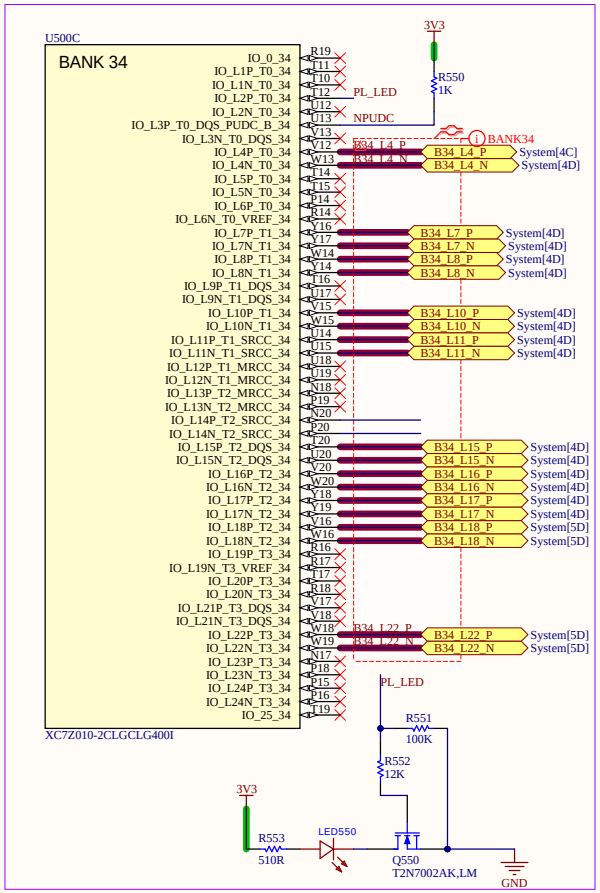
Boot Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
Signal/Mode	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[3]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[0]
Boot Devices							
JTAG Boot Mode; cascaded in most common [1]			0	0	0	0	JTAG Chain Routing [2]
NOR Boot [3]			0	0	0	1	• '0': Cascade mode • '1': Independent mode
NAND			0	0	1	0	
Quad-SPI [3]			0	1	0	0	
SD Card			0	1	0	1	
Mode for all 3 PLLs							
PLL Enabled			0	Hardware mode for PLL to lock; three instructions BootROM.			
PLL Bypassed			1	Allows for a wide PLL CLK frequency ranges			
MIO Bank Voltage [4]							
	Bank 1	Bank 0					
2.5 V/ 3.3 V	0	0	Voltage Bank 0 includes MIO pins 0 thru 15.				
1.8 V	1	1	Voltage Bank 1 includes MIO pins 16 thru 53.				

Notes:
1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
3. The Quad-SPI and NOR boot modes support execute in place (this support is always non secure).
4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU trace port. Other interface configuration may also span the two banks.

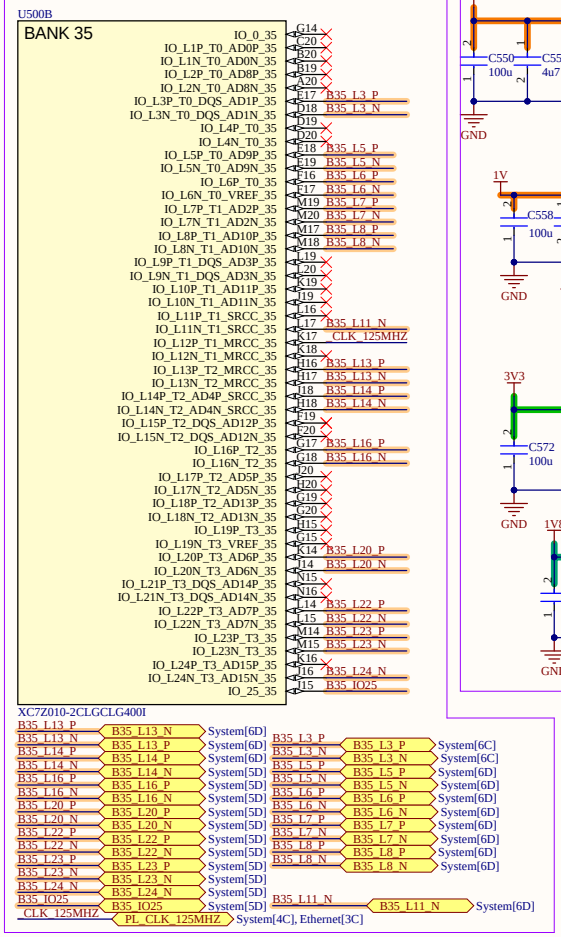
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Date: 10/27/2026		Sheet of	
File: C:\Users\..\FPGA.SchDoc		Drawn By:	

Zynq Programmable Logic

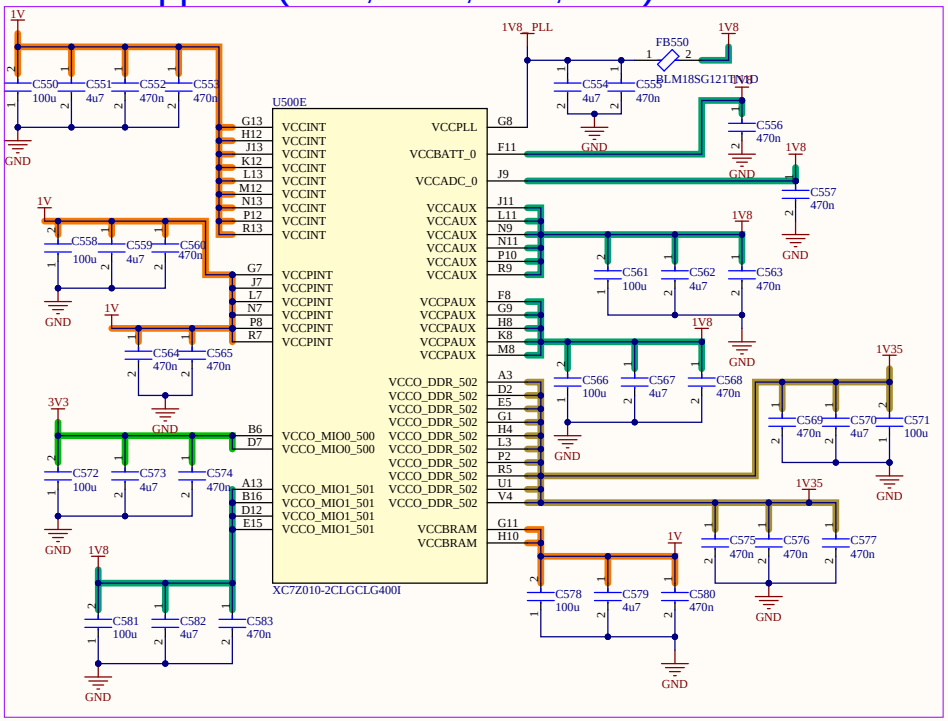
Bank 34



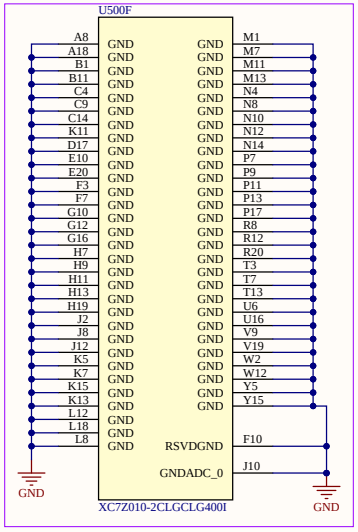
Bank 35



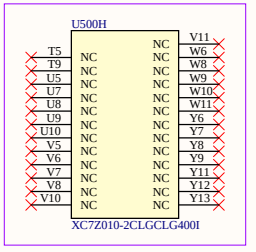
Main Supplies (AUX, DDR, INT , MIO)



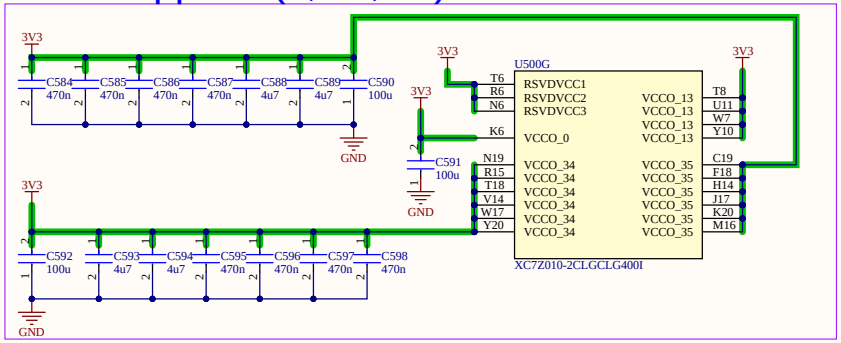
GND Connections



Unused Pins

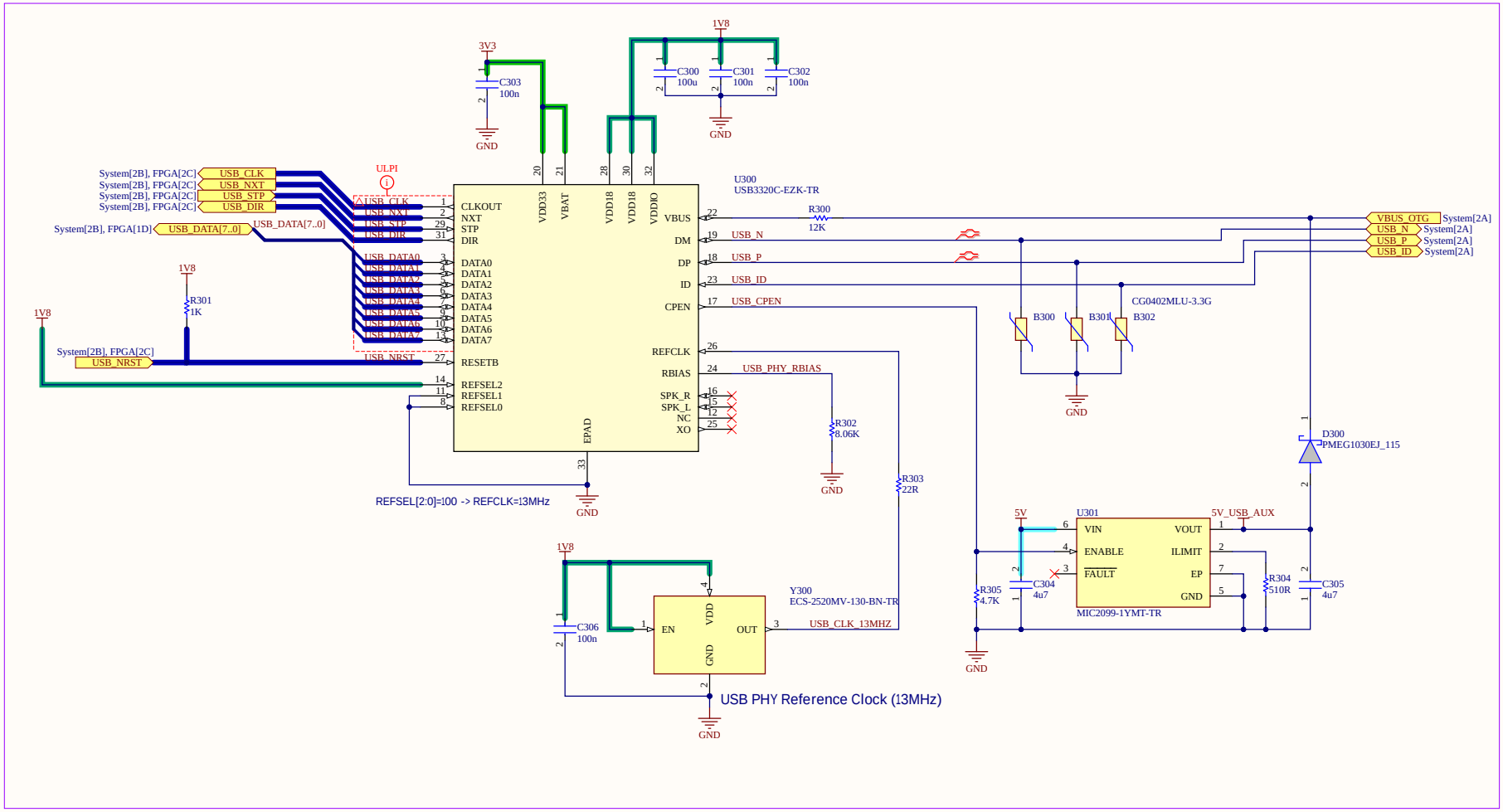


Bank Supplies (0, 34, 35)



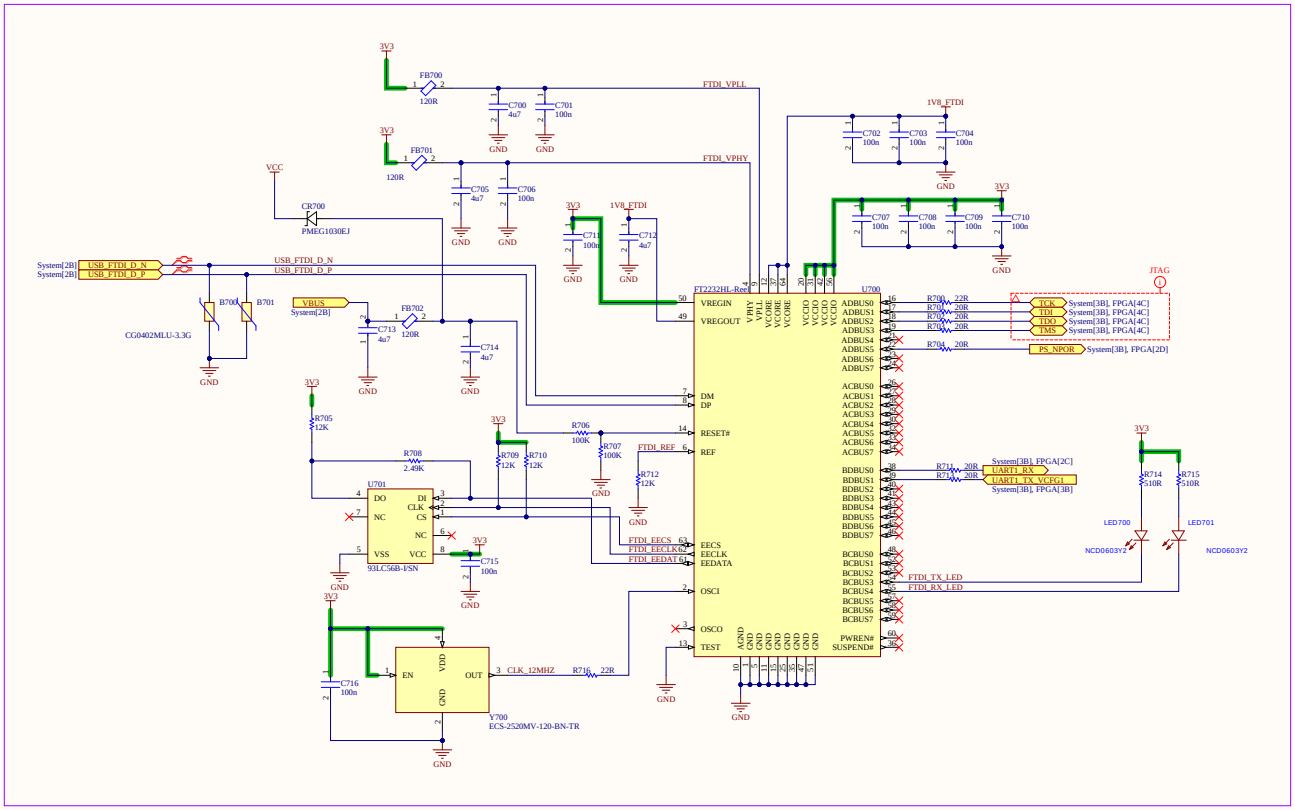
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Date: 10/27/2026		Sheet of
File: C:\Users\..\FPGA PL.SchDoc		Drawn By:

USB 2.0 High-Speed (OTG)



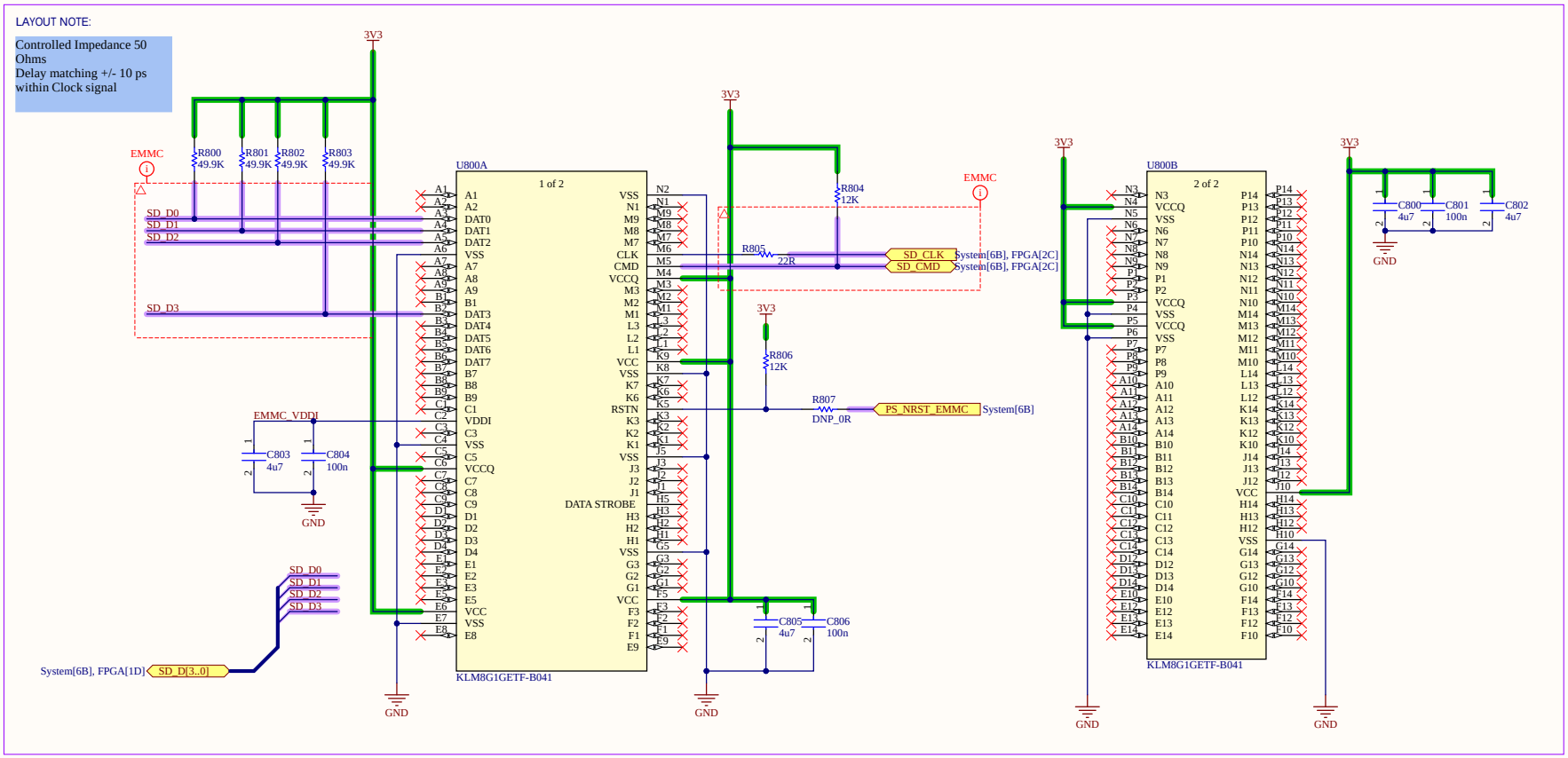
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Date: 10/27/2026		Sheet of
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USB-to-UART and FTDI JTAG Programmer



Title		
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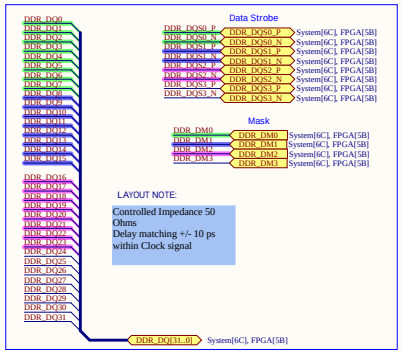
EMMC Memory



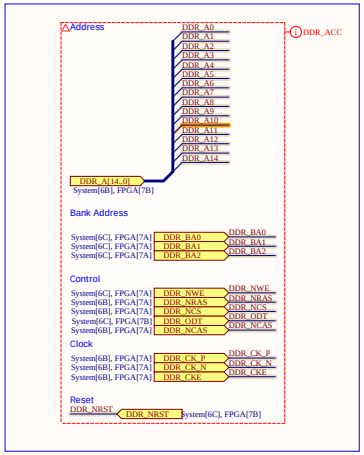
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4GB DDR3L Modules

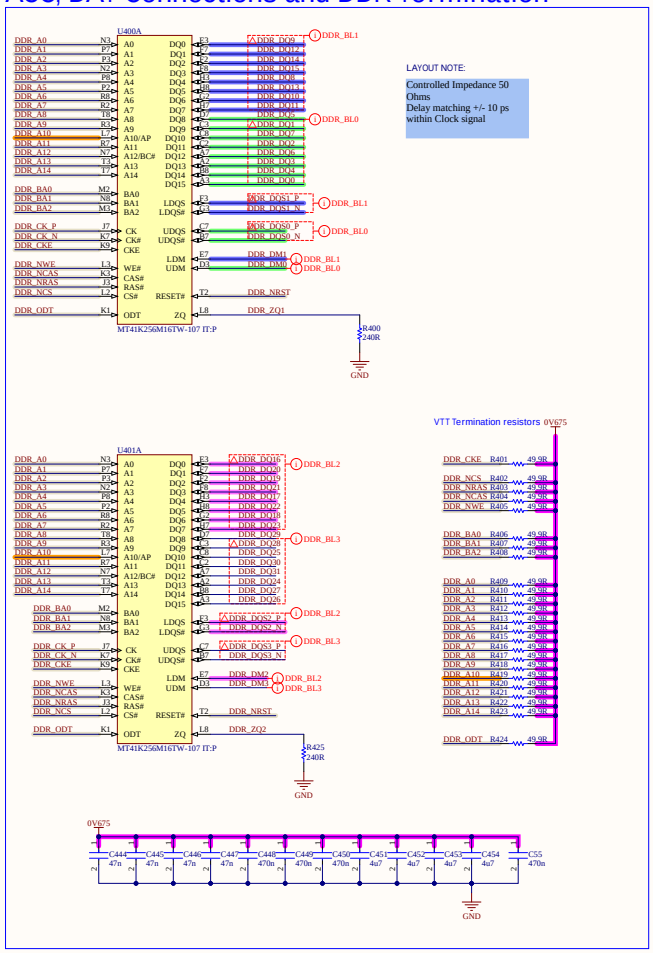
Data Signals, Data Strobe, and Mask Signals



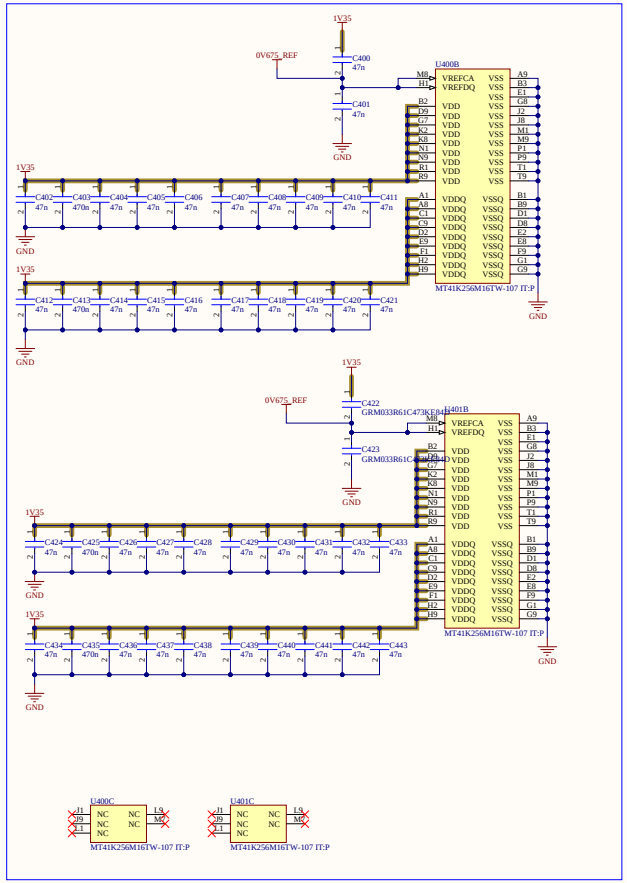
Address, Clock, Command, and Control Signals



ACC, DAT Connections and DDR Termination



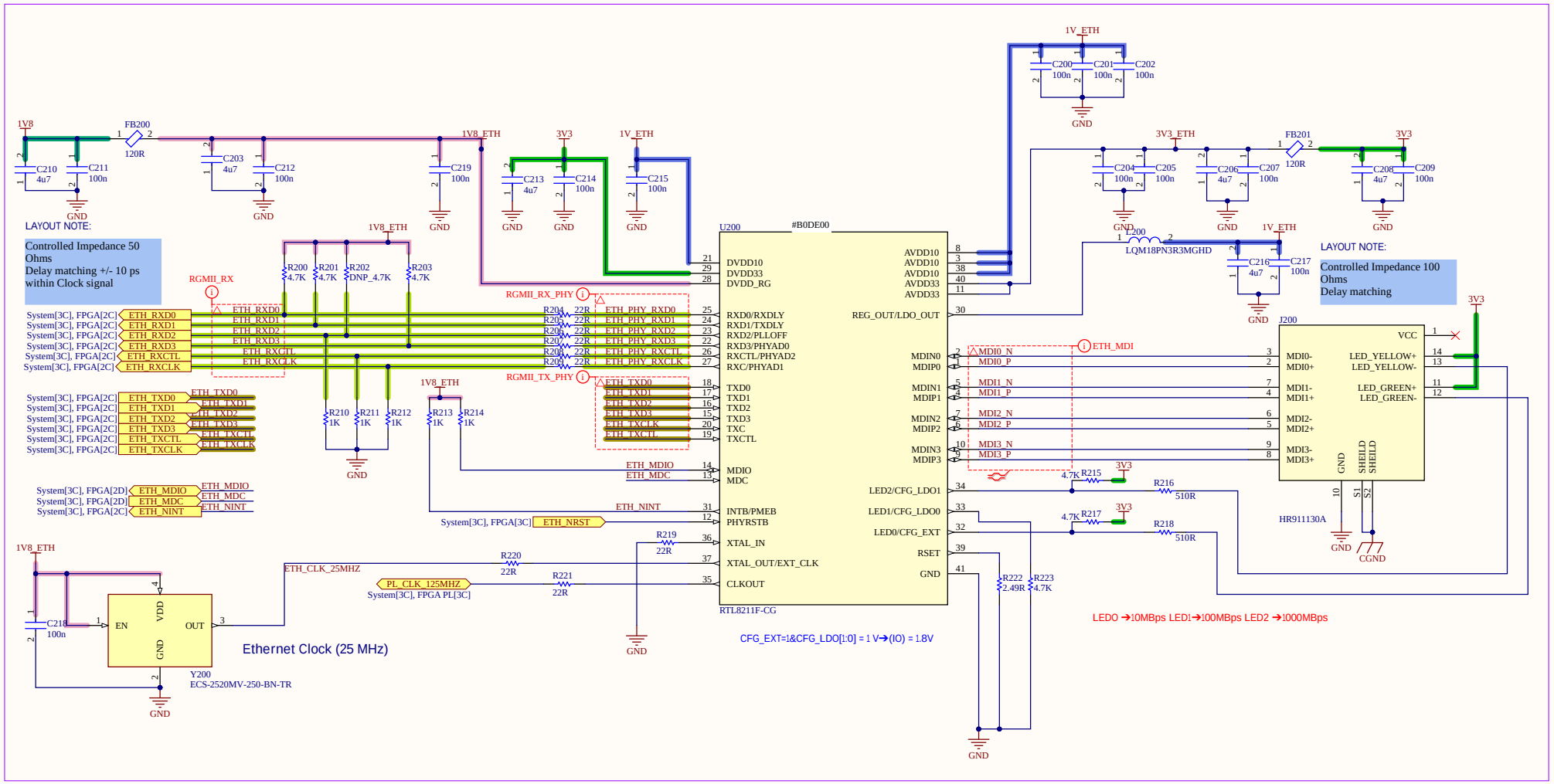
Power and Decoupling



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Gigabit Ethernet

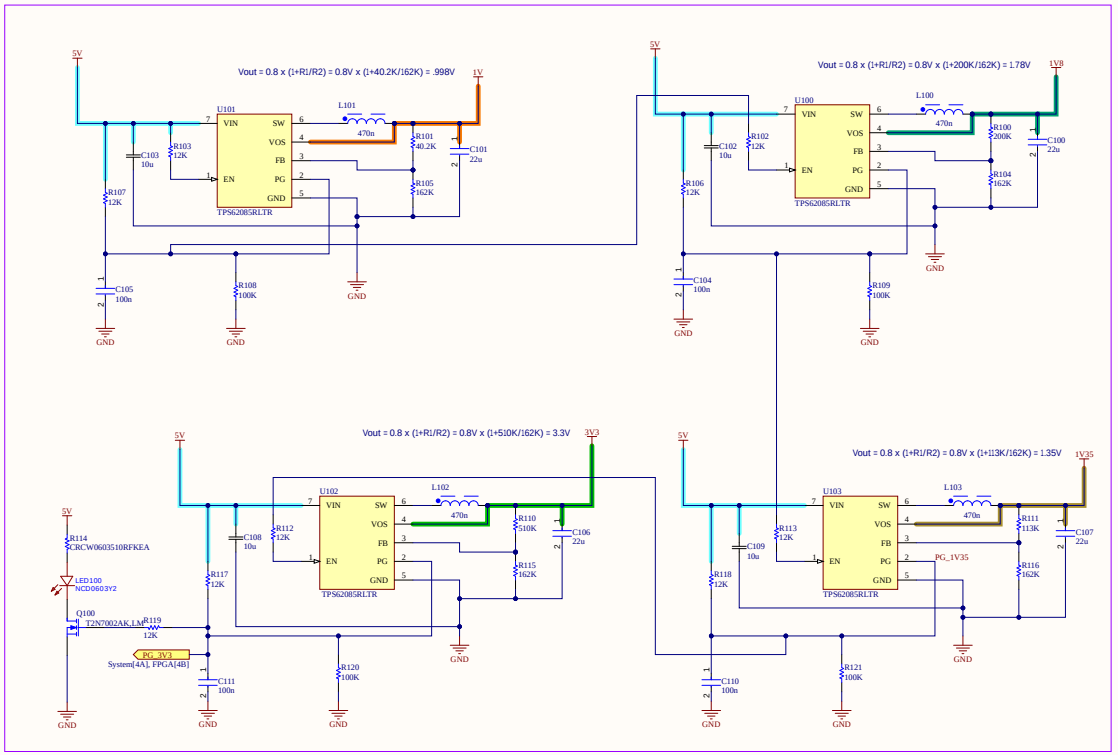
Ethernet PHY & RJ45 Interface



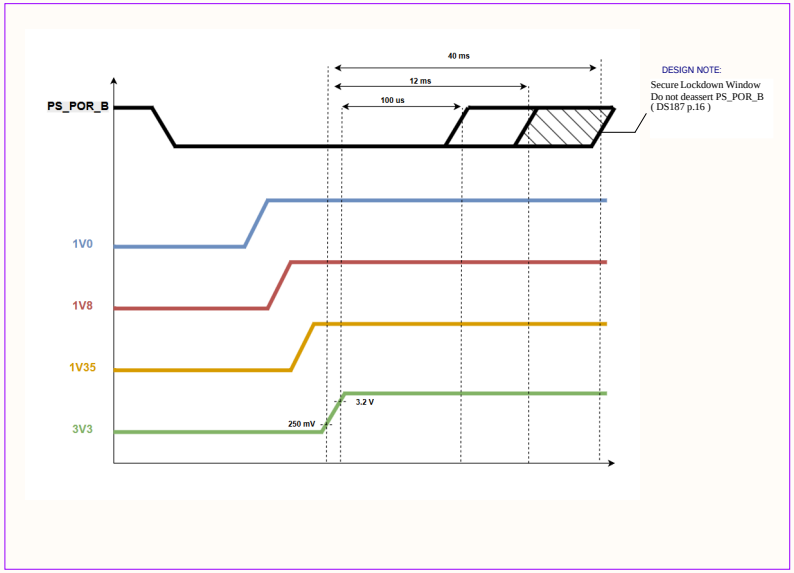
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Power supplies

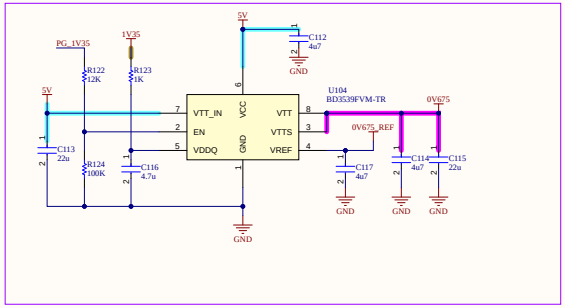
Buck converters



Zynq Power-ON sequance



DDR3L VTT Regulator



Title		
Size: A2	Number	Revision
Date: 10/27/2026		Sheet of
File: C:\Users\..\Power Sequence.SchDoc		Drawn By: